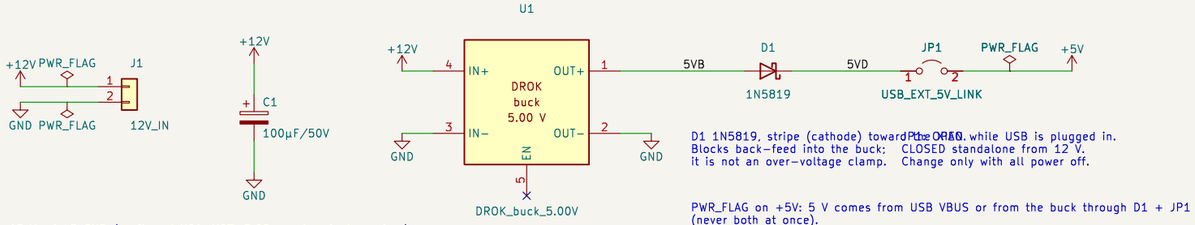
RoomCleaner winch station — Rev D, as soldered on the 5 x 7 cm perfboard

Net labels and power symbols carry the perfboard net names; every connection below is generated from perfboard_layout.py (PINS / BARE / WIRES / NETMAP). Same name = same net.

POWER: 12 V in, buck to 5 V, D1, JP1 link to the XIAO 5V pin



J1 pin 1 +12 V, pin 2 GND (perfboard X01, X03; 5.08 mm terminal or wires).
12 V comes from the supply branch: the PWR_FLAGs mark the connector as the source.
C1 sits at VMOT (perfboard T02 +, T03 −).

U1 EN is unconnected (as soldered). Select 5 V with the module's 5V pad; set and check 5.00 V BEFORE connecting loads. Non-isolated: IN− = OUT−.

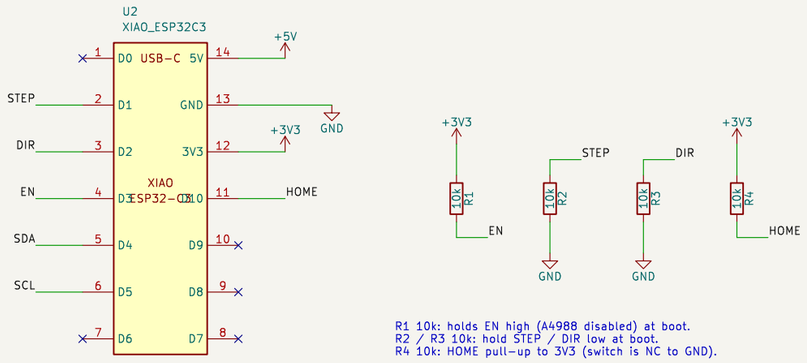
D1 1N5819, stripe (cathode) toward the OPEN. Blocks back-feed into the buck; it is not an over-voltage clamp.

JP1 OPEN while USB is plugged in. CLOSED standalone from 12 V. Change only with all power off.

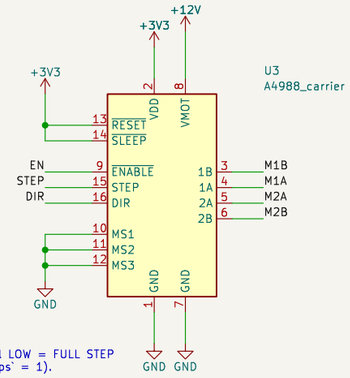
PWR_FLAG on +5V: 5 V comes from USB VBUS or from the buck through D1 + JP1 (never both at once).

CONTROL: XIAO ESP32-C3, A4988 carrier, pull resistors, motor

U2 perfboard: top row B03–H03, bottom row B09–H09,
USB-C points LEFT off the board edge.

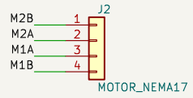


R1 10k: holds EN high (A4988 disabled) at boot.
R2 / R3 10k: hold STEP / DIR low at boot.
R4 10k: HOME pull-up to 3V3 (switch is NC to GND).



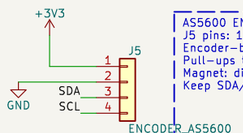
MS1 MS2 MS3 all LOW = FULL STEP (config `microsteps` = 1).

U3 perfboard: pads 9–16 in column L (rows 03–10),
pads 8–1 in column Q (rows 03–10), chip + pot UP.
Pad 7 GND = motor return (PGND on the board), pad 1 GND =
logic ground; the board ties them with wire S03–011.

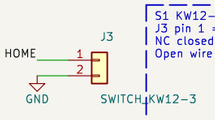


J2 MOTOR (X05–X08): pins 1–4 = 2B 2A 1A 1B. 1A/1B = coil 1, 2A/2B = coil 2.
Never plug / unplug the motor with power on.

OFF-BOARD PARTS (connectors on the perfboard, device is not)



AS5600 ENCODER BOARD (off-board), J5 C17–F17
J5 pins: 1 = VCC (3.3 V), 2 = GND, 3 = SDA, 4 = SCL. 0x36, I2C 400 kHz.
Encoder-board DIR strapped to GND and OUT/PGO open: on that board, NOT on J5.
Pull-ups to 3.3 V only; set any 3.3 V / 5 V pad to 3.3 V. Never 5 V on ESP32 pins.
Magnet: diametric disc on the spool axis, 0.5–3 mm above the chip, centred.
Keep SDA/SCL/3V3/GND < 20 cm, twisted, away from motor wires.



S1 KW12-3 SWITCH (off-board), J3 H17–I17
J3 pin 1 = NC contact -> HOME (D10 / GPIO10), pin 2 = COM -> GND. NO unused.
NC closed at rest: firmware INPUT_PULLUP + R4 10k to 3V3.
Open wire reads STOP, a short to GND can read READY: test both states.

A4988 CURRENT LIMIT — DECISIONS.md D30 (2026-10-21)
Vref = 0.85 V (about 1.06 A, at the carrier's maximum). Vref = 8 x I x Rsense, R100 = 0.100 ohm.
Supersedes D29 / Rev D README (1.35 A, 1.08 V): 1.08 V is NOT reachable on this carrier
(red R100 clones top out at ~0.79–0.85 V Vref on 3.3 V logic). VDD stays 3.3 V.
Set Vref with 12 V on, motor DISCONNECTED, logic powered, EN high: black probe on driver GND,
red on the trim pot. Read both sense resistors first. Heatsink stays on; 60 min hold test.

MICROSTEP (MS1/MS2/MS3): all three tied LOW (GND) = FULL STEP -> config `microsteps` = 1
(config/robot.example.json). If they are moved to +3V3 (1/16 step) set `microsteps` = 16 for
that station before the first run. Hardware and config MUST match: mismatch = wrong lengths.

DO NOT RAISE VDD TO 5 V. A4988 VDD, RESET, SLEEP = 3.3 V from the XIAO 3V3 pin; the ESP32-C3
GPIOs (STEP, DIR, EN) are not 5 V tolerant. Pull-ups to 3.3 V only.

FUSING IS AT THE SUPPLY END: branch fuses F1–F4 (rating TBD) and F0 7.5 A (existing, recorded,
not approved) sit before the long cables. No fuse, reverse-polarity or over-voltage protection
on this board. Rev D TP0–TP4 are not parts on the perfboard: probe the module pins / holes.

POWER-UP SEQUENCE (summary of Rev D sheet 5)

1. Remove USB and 12 V; let C1 discharge; do not short it with a tool.
2. Check the board against this netlist with a meter: both A4988 GND pads to J1–2; 12V, +3V3, 5VB/5VD/+5V each free of shorts to GND.
3. Test the supply and the buck alone: set 5.00 V at U1 OUT+ (net 5VB) BEFORE connecting any load. JP1 OPEN.
4. Add XIAO, then A4988, one at a time; power off between changes.
5. 12 V on, EN high, motor disconnected: check +3V3 and set Vref 0.85 V (D30). Never guess Vref from the screw direction.
6. Power off; connect the coil pairs (1A/1B, 2A/2B). Never plug or unplug the motor with power on.
7. Run only a short, unloaded jog and the switch test. No suspended load or room operation until bench-checked.

USB / EXTERNAL POWER

USB plugged in: JP1 OPEN. Standalone from 12 V: USB unplugged, JP1 CLOSED. Never USB + 12 V with JP1 closed (two sources meet on VBUS). Change JP1 only with all power off.

PERFBOARD HOLE MAP (generated, component side, A01 top-left)

J1: 1=X01 2=X03
J2: 1=X05 2=X06 3=X07 4=X08
J5: 1=C17 2=D17 3=E17 4=F17
J3: 1=H17 2=I17
JP1: 1=S11 2=R11
C1: 1=T02 2=T03
D1: 1=T11 2=X11
U1: 1=X15 2=W15 3=V15 4=U15 5=T15
R1: E13 – E10
R2: C10 – C11
R3: D10 – D11
R4: H13 – H16
U2: B03..H03 (5V..D7) and B09..H09 (D0..D6)
U3: L03..L10 (pads 9..16) and Q03..Q10 (pads 8..1)

Pin 1 = first hole listed. U1 EN (T15) and XIAO D0, D6, D7, D8, D9 are unconnected on the board (no-connect flags).

DIFFERENCES FROM THE REV D SCHEMATIC (perfboard wins)

Current limit: Rev D sheet 3 / README show 1.35 A, Vref 1.08 V (D29). This sheet follows D30: Vref 0.85 V, about 1.06 A.
Encoder: Rev D sheet 3 draws DIR to GND and OUT open at the encoder. The board has a 4-pin J5 only (3V3, GND, SDA, SCL), so those two are on the encoder board, not on the perfboard.
Test points TP0–TP4 of Rev D are not parts on the perfboard.
Grounds: Rev D shows one GND; the board has two A4988 GND pads (pad 7 motor return, pad 1 logic) tied by wire S03–011 = one GND net here.
Packages as built: R2/R3 stand upright (2.54 mm), R1/R4 lie (7.62 mm); J1 is a 5.08 mm terminal pitch (2.54 mm header otherwise).
Net names: perfboard 3V3 / 5V / 12V are the +3V3 / +5V / +12V power symbols; 5VB = buck OUT+ / D1 anode, 5VD = D1 cathode / JP1.

Net labels = perfboard nets; PGND and GND are one net (tied by wire S03–011)
Generated by build_kicad.py from perfboard_layout.py; DECISIONS.md D30 current limit

RoomCleaner

Sheet: /
File: RoomCleaner_Station_RevD.kicad_sch

Title: RoomCleaner winch station — Rev D (as soldered)

Size: A3	Date: 2026-10-21	**Rev: D**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1